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Computer Vision

Assignment #01

Report

1. Introduction

This project implements a human activity recognition pipeline using MediaPipe Pose. The system detects human body landmarks from video frames, computes joint angles, and classifies activities using a rule-based approach.

2. Video Source

Video File Used: cv_project_video.mp4(Download From Youtube)

The video contains activities including standing and squatting. The person remains visible throughout the video.

3. Pose Detection and Pre-processing

MediaPipe Pose was used as the pre-trained pose estimation model. The model extracted body landmarks such as shoulder, elbow, hip, knee, and ankle from each video frame. Skeleton overlays were visualized on the original video frames. A smoothing filter was applied to reduce noise and jitter in angle values.

4. Joint Angle Computation

Three important joint angles were calculated:

- Knee Angle
- Hip Angle
- Elbow Angle

The angles were computed using the geometric relationship between three body points. The angle values were tracked across all frames and plotted over time.

5. Rule-Based Activity Classification

Activity	Condition
Squatting	Knee Angle < 110 and Hip Angle < 120
Standing	Knee Angle > 150 and Hip Angle > 140
Transition	Intermediate Frames

The rule-based classifier successfully identified activities frame-by-frame using joint angle thresholds.

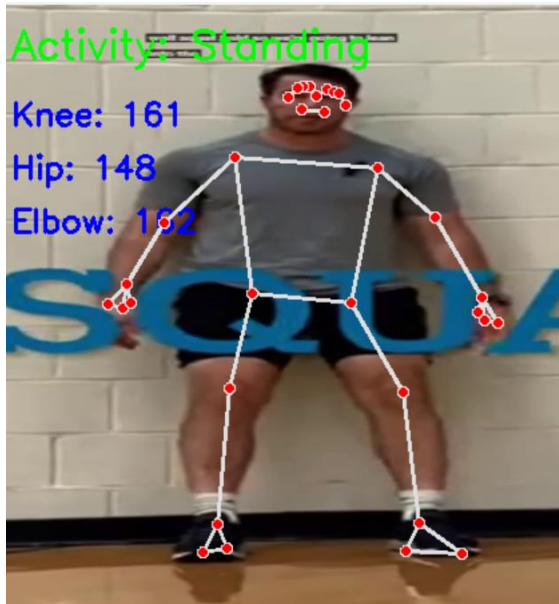
6. Accuracy Results

The activity recognition system achieved an overall accuracy of **51.12%** on the selected video. The accuracy was calculated by comparing the predicted activity labels with manually defined ground-truth labels. The results indicate that pose-based angle analysis can classify human activities; however, some frames were misclassified due to activity transitions, pose variations, and threshold limitations.

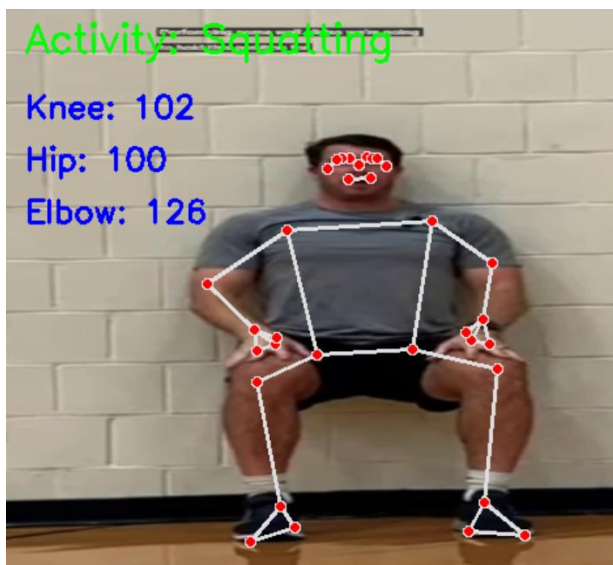
7. Observations

- MediaPipe successfully detected body keypoints in real time.
- Joint angles clearly changed during activity transitions.
- Squatting activity showed a significant decrease in knee angle.
- Smoothing reduced noisy angle fluctuations.
- Transition frames were detected between standing and squatting.

8. Standing Detection:



9. Squatting Detection:



10. Conclusion

This project demonstrated the use of pose estimation and rule-based activity recognition using computer vision techniques. The system successfully detected human poses, computed joint angles, visualized skeletons, and classified activities from video data.

11. GitHub Repository

GitHub Link: _____

